

Pseudo-Gloss Generation for Improved Gloss-Free Sign Language Translation

Student Name: L.C.Birch

Supervisor Name: A.Atapour-Abarghouei

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Abstract—Sign language translation is a new and developing field that sits at the intersection of Computer Vision, Deaf Studies and Natural Language Processing. The biggest issues in sign language translation are the lack of research into using large-scale datasets and a lack of consideration of the complexities of the sign languages being translated. Focusing on British Sign Language, we look at how we can improve the translation of the data when we have a lack of in-depth annotations and how pseudo-glosses can be used to achieve this. This work aims to develop a system of pseudo-gloss generation that takes into account the structure of the languages being translated to improve the performance of the translation model. Looking at previous work, we look at how we can improve the performance of the generation. This addresses the lack of gloss annotations in large-scale datasets. This generation of glosses can also be used for other future models to help the development of gloss-free translation.

Index Terms—Computer Vision, Machine Translation, Linguistics, Sign Language



1 INTRODUCTION

Sign Language Translation (SLT) is a complex and interdisciplinary task which often crosses multiple disciplines such as Deaf Studies, Computer Vision and Natural Language Processing (NLP) [1]. Computer vision research into sign language started very limited in scope and did not appreciate the complexities of sign languages, looking at the task of SLT as a simple recognition problem, not a translation problem [2]. The SLT field now focuses on two key areas: continuous sign language recognition, which involves identifying distinct signs within continuous video streams, and continuous sign language translation, which aims to translate these videos into their spoken language equivalents. It was only after 2018 that sign language translation began to be widely recognised as a more complex problem within the context of computer vision, due to the publication of two highly influential papers in the field that year [2], [3]. The approach to SLT was significantly impacted by the development of deep learning and neural networks enabling these models to be created. This development in computer vision has enabled the progress of research into Sign Language to develop further and sparked new development in using transformer models.

There are many sign languages throughout the world that are commonly used today, not just one unified sign language which some people assume [4]. Signed languages are their own languages separate from the respective spoken languages, usually having different word order and grammar to the spoken language [5]. Research into sign languages is a new field compared to research into spoken languages and historically people did not treat sign languages as the complex languages that they are [5]. This attitude towards sign language can be seen in some research whether intentional or not. Research in the past has not treated SLT as a proper translation problem [2], limiting the development

of the field. Along with different word order, sign languages are conveyed using the hands (manual features) and the face and body (non-manual features) [6], where a shake of the head or a facial expression can substantially change the meaning of the hand sign. Signed languages can also make use of simultaneous grammar, with two hands being able to convey different information at the same time, compared to spoken languages only using the mouth [6]. These aspects of sign language transform SLT into a complex issue rather than a simple recognition problem.

One of the main difficulties limiting SLT today is the size and quality of the datasets available. Datasets for continuous SLT can be split into two different types of datasets: datasets that have gloss annotations [2] and those without [7]. Gloss annotations [1], [5] in the field of SLT are where each sign in the video has a reference to the meaning of the sign. This will usually be the corresponding spoken language word, occasionally with some extra context about the sign. Datasets that do not contain gloss annotations usually contain the spoken language translation of the sentence, often from subtitles [8], [7], [2] due to the manner in which many datasets are obtained. Gloss-free datasets are often larger as they require less work to obtain, but are still magnitudes smaller than textual datasets used for translation [9], and so often have a wider subject area [7], making the models created using them much more relevant to the needs of Deaf people. Datasets with gloss annotations are currently able to create more accurate models but often have a restricted domain like weather forecasts [2] so they are less useful for a general-use translation model.

This paper focuses on gloss-free translation, so we can make use of these larger datasets to help create a model that can be used for a larger domain and be of more practical use for Deaf people. It is also important to note the bias that

these datasets have, with most of them solely using interpreters for their source of data, leading to less representative data. The datasets also have a lack of variation in the people signing compared to the people producing text in textual datasets. Many datasets have very few signers in their data, which can impact the training of the model.

Research into SLT so far has focused on using datasets with glosses with fewer gloss-free models being created. Due to much of the research expanding on the research by Camgoz with gloss translation [2], [3] the majority of the models have been created using the RWTH-PHOENIX-Weather 2014T (Phoenix14T) dataset created and then used in Neural Sign Language Translation [2]. This dataset is a German Sign Language dataset that uses videos of German weather forecasts and German Sign Language interpretation which is gloss annotated. This means that most of the translation models created are limited in domain scope.

Most SLT models use some form of encoder-decoder architecture, [10] often using transformers [3] for the act of translations. There are a few different approaches that models use. Models produce the output directly using the sign language video or by using an intermediary stage of the gloss representation which can then be translated into the spoken language. Architectures that do not use glosses as an intermediate stage often use them for supervision to increase the accuracy of the model. Models also consider different combinations of inputs obtained from the data. Some of these models also take into account a selection of the whole video frame [10], the face [11], the skeleton pose estimation [12] and embeddings [13] that take into account the different features.

It is important that the research extend beyond the lens of computer vision and instead takes into account other research areas. NLP is an important research area that can be used in conjunction with computer vision to increase the quality of translation [1], particularly when using glosses for translation. Deaf studies can be used to create tools to enhance the lives of Deaf people. Deaf studies can also directly inform the research done and in turn, increase the quality of SLT models. We can also learn more about the limitations of our work through critiquing it through a Deaf studies lens. This research focuses on British Sign Language (BSL) to English translation so it is key to look at the properties of BSL to inform the translation. BSL has many features, shared with other sign languages, that make it a complex language to translate - pointing [14], variation in signs used [15], [16], regional signs [5], [17], [4] and variation in signing structure [5]. The variation in regional signs in particular is thought to be present in BSL due to the various residential schools in the UK and the banning of sign languages in said schools [15]. Due to the circumstances leading to the development of BSL, it has a significant amount of variation making this a particularly important consideration.

Using this multi-disciplinary approach, the research question is "How does using improved pseudo-glosses impact the performance of a gloss-free sign language translation model?". The project aims to improve the performance of SLT models by improving the generation of pseudo-glosses [9], working towards being able to generate pseudo-glosses that are closer to real gloss annotations for large-scale datasets. Previous work on generating pseudo-glosses

either uses current gloss annotated datasets to create a machine learning model [18], rule-based generation [19] or a combination of both [9], all using the Phoenix14T dataset. This work will build upon prior research, exploring how existing approaches can be improved and applying it to large scale datasets.

The focus of improving these glosses will look at two different perspectives, improving current rule-based systems by improving clause segmentation methods and the use of Large Language Models (LLM) in generating pseudo-glosses. This builds upon the widespread usage of LLM in NLP and the prior research using LLM in the specific domain of SLT. For the improvement of clause segmentation we look at syntactic parsing and how this can be applied to find the boundaries between the clauses and how the sentence is built up. Using this contextual information about the English sentence, we can use this to help apply our rule-based system more effectively. Specifically, for the usage of LLM we look at prompting the LLM to generate pseudo-glosses by instructing it with context about the Structure of BSL and the typical word order and structure of sentences namely the topic-comment structure. We propose to use this for pseudo-gloss generation to support existing large-scale gloss-free SLT datasets. These will be used to enhance the performance of the model, trying to accomplish a similar performance to gloss-annotated datasets. Glosses can help the model learn the order in which signs are represented in the data. This system aims to help future work with similar datasets, as it can be modified for datasets to produce similar annotations without expensive gloss annotation.

The model created will be trained on the BBC-Oxford British Sign Language Dataset (BOBSL) dataset [7] due to it being the largest BSL dataset available. This dataset has limitations due to the large amount of variation in BSL in regional signs around the country and the natural variation of sign language (for example due to age and cultural background) not being represented [7]. The dataset is also obtained from broadcasts of interpreted footage, which limits the usefulness of this data. Organic BSL is different from interpreted BSL, which means that if the model were used on Deaf people in the world there could be limitations in performance.

There are also general limitations of current research so far in SLT, with no model being created that can create consistent, high-quality translations. The current highest-performing models are unable to accomplish this with limited datasets [10], so applications for Deaf people are limited. The datasets used also limit the generalisability due to sample bias and the limited variation in the setting.

2 RELATED WORK

This section will look at the state of sign language technology research thematically. We have decided to look at the state from an SLT perspective as this is the focus of the paper. The subsections in this section are Deaf Studies and Interdisciplinary Work, Sign Language Recognition, Gloss-annotated Sign Language Translation, Gloss-free Sign Language Translation and Datasets. A focus on Deaf studies is not typical for sign language technology research but the decision has been made to include this as it is a major gap

in a lot of the research so far and can help inform how we approach SLT.

For all computer science-focused research, we limit our scope to work published from 2018 onwards, as this marks the introduction of deep learning to SLT. Research prior to this is considered less relevant to SLT as a deep learning problem

We did not include Sign Language Production due to its limited relevance or SLT that does not use a video for input, excluding the use of sensors [20], [21] and gloves [22]. The Deaf community does not support this approach and they prefer a less intrusive approach [6].

2.1 Deaf Studies and Interdisciplinary Work

Even though the main focus of this paper is computer vision research, a look at relevant Deaf studies research can be key to understanding the limitations of current SLT models and how we can improve them. Most of the Deaf studies research done is not directly related to sign language technology however, there have been a few interdisciplinary papers that can directly inform how we can improve research.

One paper has been written on best practices for this research from a Deaf studies perspective [6]. Fox, 2023 examines the work that has been done in sign language technology so far and the shortcomings of this research through the lens of Deaf studies. It identifies five key issues that are common in research - knowledge of the sign language translated and general knowledge about sign language; involving Deaf people in research, as participants and researchers; motivations of the researcher and their relationship to the sign language community; the sign language data to be considered and recognition of the limitations and challenges of sign language technology research. Some of the points made in this research are currently looked at in some active SLT research [2], [1], [23], particularly a basic knowledge of sign languages, consideration of the dataset used and involving Deaf people in research. However, as the paper sets out there are actionable points that could improve the quality of research that are not done, particularly knowledge about the specific sign language being translated and a greater consideration of the reasons for the research. Even the points that are considered are not considered consistently in SLT research.

This paper has shortcomings when looking at the papers it draws from, many of the papers referenced are research from before 2018 which was a turning point for SLT research. It was the time of the first major application of deep learning and when SLT was considered a complex problem. Whilst the criticisms are still valid as many papers to this day still have similar issues it takes a less optimistic stance on how SLT research can aid communication between Deaf signers and non-signers. This is backed up by an article from 2016 [24] so it is hard to determine the validity of their claims due to the changes of the landscape of computer vision.

Another paper which looks at Sign language technology research through an interdisciplinary lens is Bragg 2019 [25]. This takes a different approach by gathering a diverse group of experts within computer science - along with researchers

from linguistics, education, psychology and Deaf studies and attendees from a tech company and asking them to discuss the problems in sign language technology. The paper looks at the current landscape, the field's biggest challenges and calls to action. The calls to action are the key points of this paper and are - involve Deaf team members throughout, focus on real-world applications, develop user-interface guidelines for sign language systems, create larger more representative public video datasets and standardise the annotation system and develop software for annotation support.

The calls to action - involving Deaf team members throughout and a focus on larger more representative datasets are common to both papers and something key to progressing SLT research. The standardisation of an annotation system is an idea with merit, but in the paper, they do not realistically look at the benefits. "A standard system would reduce annotation cost and errors" whilst true would not drastically benefit SLT research as the datasets needed are so large that even cheaper annotation would not drastically benefit the research. A standard annotation system would not help as many large-scale datasets are simply not annotated by hand and just draw on the already available information. A focus on real-world applications can be helpful, particularly for the collection of datasets and any models that are going to be used in the real-world, but there is merit in work that is less closely related to real-world applications. Algorithmic improvements and different approaches to data embeddings can be brought to real-world applications in the future and the paper fails to highlight this.

Using this research we should focus on three things to enhance the development of our SLT models. Models should take into account non-manual features along with manual features and take into consideration how they will use them for translation. Glosses can be very useful in increasing performance but how we try and use them to increase performance should be considered. Models using them as an intermediate stage should consider if other methods would work better since glosses are an inherently restrictive method of representing Sign Language. Word order should be considered since some models do not sufficiently explore the differences in word order between spoken and sign language. Models having more explicit knowledge about word order, particularly in gloss-free datasets could help performance instead of letting models learn from the datasets, with this approach it is also important to consider that not all sign languages follow the same sign order, something which is not considered by SLT literature generally. There are many other ways how you could use Sign Language Linguistics to inform your work but this focuses on those that are the most important and often overlooked in models.

2.2 Datasets

Datasets used for sign language technology research can be split into three different categories: isolated datasets, continuous gloss-annotated datasets and continuous gloss-free datasets. They have importance as the quality and size of the data directly impact the models performance. We will look at only the continuous datasets as isolated datasets have limited real-world use.

The most widely used dataset is the Phoenix14T dataset [2]. This is a publicly available dataset taking videos of weather broadcast interpretation footage. It contains 11 hours of videos, a sign vocabulary of 1066 signs 76K sign annotations and 9 signers. The specific domain means that the dataset is not very applicable for real-world use and the size of the dataset is very small for a translation model. A lack of diversity in the signers hinders the generalisability of the model and the method of gathering the data means all of the data is translated from the spoken language which impacts the quality of the data. The Gloss annotations available do help with performance and the widespread use of the model makes it good for comparison.

The largest BSL dataset is the BOBSL dataset [7]. This is another dataset taking TV footage of BSL interpretation as the source. It contains 1467 hours of videos, a vocabulary of 2281 signs and 39 signers. It lacks organic sign language due to the data being interpreted data but has more signers than other broadcast datasets and has a wider domain and is one of the largest sign language datasets.

There are other large-scale datasets, openASL [26] is an American Sign Language dataset which contains videos collected from online video sites such as YouTube. It contains 288 hours of videos with over 200 signers. The organic nature of the dataset along with signer variation is positive but this datasets lacks validity of the quality of the sign language data as many of the data could be of learner signers and have bad quality annotation.

Further large-scale datasets with more organic sign language in real-world environments would significantly progress SLT and SLR research. A more diverse group of signers for further datasets is also important to help models deal with variability in sign language. These actions would help work to be more applicable in the real-world.

2.3 Sign Language Recognition

Sign Language Recognition is the act of being able to recognise sign language glosses and can be split into two categories - isolated SLR (ISLR) and continuous SLR (CSLR). ISLR is the act of looking at videos of individual signs and determining what sign it is. It is a less complex problem with few applications to SLT or the real-world. CSLR is the act of being able to recognise the sequence of glosses in a video clip of sign language. It is sometimes used as a subtask for SLT when using gloss-annotated datasets using a sequence of glosses as information to translate into the spoken language. There are also other applications for the Deaf community outside of translation in particular searching sign language videos by sign is a potential application for recognition [6].

CSLR as a problem goes back further than SLT and is more popular in research despite its limited applications. CSLR models usually follow an encoder-decoder structure. The encoder is used to extract features from the Sign Language Video and the decoder is used to generate a gloss sequence using the features obtained from the encoder. We can use different neural network structures and "previous models have used different Convolutional Neural Network structures for the encoder - 3D-CNNs [27], [28], 2D-CNNs [29], [30] and 2D+1D CNNs [31], [32] with papers using a mix of single-stream and multi-stream architectures" [10].

One of the current highest performing models is the Two-Stream Network for Sign Language Recognition and Translation [10]. Which achieves state of the art performance and uses CSLR as a subtask for translation showing the use of CSLR in SLT. This paper looks closely at the encodings for recognition and uses two different streams with different encodings. One stream encodes RGB images and the other stream uses keypoint estimation - focusing on both hands, face, and upper body keypoints. The keypoint estimation stream is used as supplementary information to the RGB stream due to the blur that decreases the accuracy of the keypoint estimation. A CTC decoder is used to generate the gloss sentences from the encodings.

The performance of the two-stream model is currently the best on the Phoenix14T dataset but does lack proper exploration of the real-world uses of SLT. It does not acknowledge the limitations of the dataset used, nor does it acknowledge the potential linguistic differences between Sign Languages that could impact the performance of the model on different sign language datasets.

2.4 Gloss-annotated Sign Language Translation

The majority of papers on SLT focus on datasets that are well annotated containing glosses [33], [13], [34]. This means that models are often designed to use gloss annotations to improve performance but are not transferable to larger datasets that do not contain extensive annotations.

The first key paper that covers this is Neural Sign Language Translation [2]. This is one of the papers that acknowledges a lot of the shortcomings of previous research. It contributed a lot to the state of SLT, being one of the first papers exploring the video to text SLT problem, releasing one of the first continuous SLT datasets, RWTH-PHOENIX-Weather 2014T (Phoenix14T) and contributing a baseline model for future papers to compare to. The paper proposes to use an encoder-decoder architecture for this task, exploring in depth different choices you can make for the architecture. Future models have consistently used some variation of an encoder-decoder architecture for SLT. This paper however does not address the limitations of the dataset chosen - the limited domain only looking at weather forecasts, limited signers and no organic sign language present nor does it explain the choice of metrics in depth. The paper was a good starting point for other work to progress from and most of the work in SLT uses the Phoenix14T dataset.

Camgoz 2020 [3] works on the previous research done in his previous paper [2] by looking at both multi-channel approaches and using a transformer architecture. It identifies a lack of focus on the non-manual features present in sign languages in the SLT research at the time and looks at how we can include the extra features to enhance the performance of the model. Previous models use a gloss intermediate stage but this limits the information available as gloss recognition does not necessarily convey all of the information, relies on the quality of gloss annotation and is an information bottleneck. It uses gloss prediction for an attention mechanism for the Sign Language Transformation Transformer and so uses the gloss annotations but not as an intermediate step to increase performance.

One of the current state of the art models is the Two Stream Network for Recognition and Translation. This does

not use an approach like Camgoz 2020 [3] and uses an intermediate gloss stage, then an MLP and a translation network which takes the gloss sentences as input outputting the spoken language sentence. This achieves the best results but again lacks an interdisciplinary perspective, doesn't acknowledge the limitations of the dataset used or the lack of real-world applications and the model is unable to be applied to gloss-free translation.

2.5 Gloss-free Sign Language Translation

There has been a recent trend to try and work towards gloss-free SLT [35]. Larger datasets for Sign Language are not annotated in-depth and often use the spoken language translations without glosses. These larger datasets [7], [8] have a more varied domain and so can produce models that are more applicable to the real-world and future large-scale datasets likely will not have gloss annotations. SLT needs to move away from commonly used datasets like Phoenix14T as they are limited in domain and size. The datasets that can create models with real-world uses will most likely not have gloss annotations so models created need to be able to deal with gloss-free data.

Looking at the research done using the BOBSL dataset [7] there are a few models that have been created - a baseline model from the original paper, a model using context [23] and a model using unsupervised learning [36].

The first model created for the BOBSL dataset [7] as a baseline model uses a standard Transformer encoder-decoder architecture. It uses features extracted from an I3D model as input and achieves significantly lower scores on the SENT-TEST subset of the data than models used for Phoenix14T, showing how much of a difference the different dataset impacts the performance of SLT models.

Progressing from the baseline model, Sincan 2023 proposes using context to improve the performance of SLT transformers [23]. Similarly to the baseline model it uses I3D features and along with this uses context via word embeddings from the captions in the dataset and spotting word embeddings from the I3D features. Each of these 3 embeddings are used in individual encoders with the output being used as the input for various decoders and for attention in the transformer decoders.

The latest model for the BOBSL dataset is from Guo 2024 [36]. This takes inspiration from work in unsupervised neural machine translation and applies it to SLT. It creates both unsupervised and supervised models. Again, this model uses an encoder-decoder architecture but does not use a transformer, unlike other work in the BOBSL dataset. This work achieves lower scores than [23] but not considerably lower. These results show that competitive performance is achievable outside of using a transformer encoder-decoder architecture.

This paper does display a systemic issue in SLT research which is a lack of understanding of the sign language being translated in this case BSL. The paper states "In the context of British Sign Language, presenters may adhere to a more English-like word order" due to the statement in Sutton-Spence and Woll 1999 "signing is particularly influenced by English word order when the signers sign while translating from a text" [5]. The differences in influence in sign due to

translation do not manifest in sticking to a more English-like word order. Translations are different from organic signing and so have significant differences in structure. However, they do not have a particularly English-like word order and this shows a lack of understanding of the structure of sign language. Looking at the unsupervised approach, the approach does not achieve good results for translation which does not show promise for using this approach in further work.

3 METHODOLOGY

The lack of high-quality gloss annotations is one of the potential factors limiting performance for gloss-free SLT models. Pseudo-glosses [18] are a way to generate our own glosses that can be used when we are lacking our own glosses.

Previous work uses the Phoenix14T dataset, which is very limited in scope, only including weather broadcasts. Previous pseudo-glosses are therefore not necessarily scalable to larger datasets, particularly when models to generate them are reliant on the annotated glosses within the dataset.

To create pseudo-glosses for our large scale dataset we utilise a rule based system from [37], which are a set of rules for translating English to BSL. Instead of translating to BSL, we can use these rules to generate pseudo-glosses in place of traditional gloss annotations. Traditional glosses are expensive and time consuming to produce, requiring expertise in the target sign language. Spotting methods are an additional method of generating gloss like features where they look at using dictionary datasets or lipreading models to determine signs that are present in the data but are reliant on other models or datasets and can be computationally expensive. Producing our own pseudo-glosses gives us another way of generating features that can be used as part of a dataset. This means we can make better use of large-scale datasets with a larger domain than the current widely used gloss datasets.

We then use these pseudo-glosses within the SLT architecture to create our own English translations looking at how we can use both the glosses and the BSL features using Transformers.

In this section, we look at how we can generate these pseudo-glosses, how the architecture should be designed around this and how we can implement and evaluate this system. These will enable us to answer how pseudo-glosses impact the performance of SLT models.

3.1 Datasets

There are many datasets that you can choose from to use for Sign Language Translation. With previous knowledge of BSL, this work focuses on BSL datasets which narrowed down the selection to four main datasets - BOBSL [7], BSL-1K [8], BSL corpus [38] and SeeHear [39]. BSL-1K is a dataset broadcast subtitles with the interpretation of the audio to get the data. BSL-1k has 1060 hours of video along with weakly aligned subtitles and has a sign vocab of 1064 signs. BOBSL is a similar dataset drawn from the same source distribution with no overlap between episodes but with significant overlap between signers and shows. The BOBSL dataset was chosen over BSL-1K as it is a larger dataset

having 1,467 hours of data covering 2,281 signs. Having a larger dataset is a clear advantage with the source of the dataset being the same, the only key difference is the size of each. BSL-1K is also not publically available so would not be a possible choice. The other two datasets have key advantages, the BSL corpus [38] is a dataset built from dialogue recorded in a lab environment from Deaf people which is gloss-annotated with 125 hours of data containing 5000 signs. The dataset also contains 249 signers compared to 40 for BSL-1K and 39 for BOBSL. The SeeHear dataset is a dataset obtained from the long-running UK BBC show SeeHear which is a Deaf TV show and so is presented using BSL. This means that there is a large variation of BSL and a large variation in setting with more than 1,000 signers and it is 90 hours long. Table 1 below shows the datasets compared to each other. The main drawbacks of BOBSL are the lack of gloss annotations and the lack of organic BSL which are both addressed by these datasets. However, they are significantly smaller and for the use of a deep learning model are not appropriate and were not selected. Previous models have been created using the BOBSL dataset, so there has been other research we can compare our results to however, this is still limited compared to other non-BSL datasets.

TABLE 1: Comparison of BSL Datasets

Dataset	Hours	Gloss Annotations	Vocab Size
BOBSL [7]	1467	39	2,281
BSL-1K [8]	1060	40	1,064
BSL Corpus [38]	125	249	5,000
SeeHear [39]	90	1000+	N/A

3.2 Evaluation Metrics

For the task of translation, the SLT community takes many commonly used quantitative metrics which are used in translation research by the NLP community as they are a standard method of computing the quality of translation. Specifically, most of the models created use BLEU [40] and ROUGE [41] metrics and so these will be used to evaluate our model. The only other metric that has been used by papers using BOBSL is the CHRF [42] metric so we will also be using this metric so that we can better compare the results to prior research.

The BLEU metric [40] was proposed to help quantitatively determine the quality of the translation using the word n -gram overlap between the output and reference translation, in this case the subtitle text. It is defined as

$$\text{BLEU} = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right), \quad (1)$$

where p_n is the modified precision for n -grams, w_n is the weight (typically uniform), and BP is the brevity penalty to penalize short translations. BLEU is effective for evaluating surface-level overlap but can penalise semantically similar sentences with different words.

The ROUGE metric [41], often in an SLT context using ROUGE-L looks at the longest common subsequence to determine quality.

$$\text{ROUGE-L} = \frac{(1 + \beta^2) \cdot \text{LCS}}{\text{Reference length} + \beta^2 \cdot \text{Candidate length}}, \quad (2)$$

where β is used to balance recall and precision set to 1.2 to slightly emphasis recall over precession.

The CHRF metric [42] looks at character sequences using character N -grams as opposed to word N -grams. The CHRF score is computed using the F_{score} formula:

$$\text{CHRF} = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}, \quad (3)$$

where β emphasises recall over precision which is set to 2.

The main drawback of these translation metrics is that the quality of translation is subjective and a quantitative approach will not give you all the information about the quality. They are also reliant on the data provided to check the quality of translation. Particularly when looking at alternative translations these metrics are not always able to determine the quality, an alternative translation can be just as accurate as the reference translation. To increase the validity of the results of our model we will be qualitatively examining the output. We will look at the translations compared to the corresponding subtitles and look at the similarity of the content, the flow of the sentence and the semantic similarities between the translation and subtitle. This is important to compare low quality translation where they could both be incorrect but some have a more natural flow which is not examined by the quantitative metrics.

3.3 Embeddings

When using deep learning models there are different ways we can deal with the data. Previous research uses frames and/or different embeddings for the input to these SLT models and they can have a big impact on the quality of translation. Using knowledge of BSL, it is important that any embedding we use contains information about the handshapes, facial expressions and body. This ensures that the model takes into account both manual and non-manual features.

For the BOBSL dataset Two-Stream Inflated 3D ConvNet (I3D) embeddings [43] are readily available for use as part of the dataset. These are a 1024-dimensional feature sequence extracted from the model trained with the Sign-Train subsection of the dataset on sign classification for a vocabulary of 2,281 signs applied with a sliding window of stride 4 [7].

These I3D embeddings capture spatio-temporal features, making them well-suited for SLT tasks through a two-stream approach that leverages both RGB frames and optical flow from the video. Having spatial-temporal features means that both the individual frames and signs within them are encoded along with the movement of the video via the optical flow. This means that all of the key features of sign language will be encoded within these embeddings since the whole frame is used. This is crucial as we need any embeddings to include non-manual features and signing.

Alternative embeddings could be used that specifically focus on sign language features like facial expressions and mouthing patterns, key non-manual features used in BSL.

Due to the I3D features being utilised in previous work, we can closely determine the impact of the pseudo-glosses since that is the key differential between the models, not the features used. One particular advantage of using the I3D features over developing our own is the quicker development time, with the features already being created as part of the dataset. This enables us to spend more time on training models. Another alternative is to use the frames and convert them into tensors. However, this approach has a potential disadvantage compared to learned embeddings, as embeddings learn to focus on the important parts of the image, whereas the model would have to learn this on its own. With limited training time, this could be a significant disadvantage to performance.

The use of these embeddings has an additional advantage with development due to the processing being completed meaning less time is needed to create additional embeddings.

With the multimodal nature of the task, we also need to deal with the textual information format for glosses. These glosses need to be encoded into their own format, which is used for the gloss encoder. As in [23] we tokenise the gloss sentences and spoken language using a pretrained BERT model [44], specifically the BERT-base-cased model from the Huggingface transformer library using WordPiece tokenisation [45].

3.4 Pseudo-gloss Generation

For the use of pseudo-glosses in the model, we need a dataset of generated pseudo-glosses which we looked at generating using a rule-based system from [37]. The current system proposed in [37] uses a rule-based system to find the word order of glosses, which are then used to find the corresponding sign in a dataset of sign language videos to create a BSL video translation from English. Part-of-speech (POS) tagging involves assigning a grammatical category, such as a noun or verb, to each word in a sentence. This is done for each sentence being translated. Each token is then ordered based on rules based on the result of the POS tagging. The order which the words should be ordered in can be seen in Table 1 from [37].

Additionally, there are extra rules that do not help with the creation of pseudo-glosses, which are not utilised in our Python implementation. Changing months into the BSL form of OCT, only looking at words that are in the bank of videos, both do not need to be considered since generation of sign language videos is not needed for this application. Currently, the system uses connectives to split up the sentence to then use the rules on each clause, which is not accurate for every clause. This is an area of improvement that we explore to improve accuracy.

For the further improvement of this rule-based system we looked at two different approaches - Large Language Models (LLM) and syntactic parsing.

3.4.1 Large Language Model Pseudo-Glosses

For the use of LLM, we look at both directly generating glosses from an LLM or using a LLM for the task of sentence splitting. With the success of LLM in sentence segmentation

[46], a similar task, we can use them for clause segmentation as an alternative to using connectives as a method of splitting.

The LLM used was TinyLlama [47]. It was chosen since the smaller size means that it is able to process prompts quicker, which is important when generating a high quantity of prompts needed to generate the pseudo-glosses. TinyLlama is a competitive model [47] compared to models with a similar number of parameters, making it a good choice for models that are faster at inference.

The first way that the LLM was used was the direct approach, using knowledge about the structure of BSL to accomplish this. BSL utilises Object-Subject-Verb order. Using this, we are able to prompt the LLM to reorder the words into that structure. The prompt used was

Convert this English sentence to sign language gloss notation:

Rules for sign language gloss:

- Reorder the words into OSV (Object-Subject-Verb) order.
- Remove any words that do not appear in BSL.

The output should only include words from the original sentence.

English: text

Sign Language Gloss:

Another feature of BSL is the many words in English that are not present in BSL. So as seen above the prompt instructs the LLM to remove any words that are not present in BSL. This prompt had a list of words that do not appear so the LLM did not have to rely on its own knowledge of BSL which could be limited.

The second way was to use both the rule based system and LLMs conjointly to generate the glosses. Firstly, using the LLM to split the sentence up into its clauses and then applying the rules from [37]. The prompt used to split up the clauses was -

Split the following English text into grammatical clauses. Use ONLY words from the original text—do not add, change, or remove any words. Return only the clauses separated by ‘—’ without any explanation.

Text: text

Clauses:

When there is an issue with the output due to the LLM hallucinating instead of using the LLM output, the system will fall back to using only the rule based system.

Due to the limited size of TinyLlama, the accuracy of the pseudo-glosses may be significantly affected by the model’s limitations rather than the methodology itself. To address this, we generated pseudo-glosses for the comparison sample using ChatGPT-4o Mini [48], a model that performs strongly on GPT benchmarks. We decided to test the direct method generating the pseudo-glosses only using the LLM and no rule-based system. This is a harder task than clause segmentation and could be have significantly higher performance on a more complex model. The prompt used for generating the pseudo-glosses was:

As a Sign Language Gloss Generator, create British Sign Language glosses for each sentence by reordering the

TABLE 2: Word Classes and Corresponding POS Tags

Word Class	POS Tags
Interjections	UH
Temporal expressions	-
Determiners	DT
Prepositions	IN
Adjectives, Numbers, Possessive pronouns	JJ, JJR, JJS, CD, PDT, PRP\$
Nouns	NN, NNP, NNS, NNPS
Foreign words	FW
Verbs, Adverbs	VBD, VBG, VBN, VBP, VBZ, VB, RB, RBR, RBS
Existential there, Modals	EX, MD
Pronouns	PRP
Question words	WDT, WP, WP\$, WRB

sentence into a topic-comment structure and removing any words not present in British Sign Language.

3.4.2 Syntactic Parsing Pseudo-Glosses

The second approach was to use syntactic parsing to split up the sentences into their individual clauses. This is to then use the rule base system on each clause instead of using connectives as the main splitting method. For this we looked at using both dependency parsing and constituency parsing.

Dependency parsing looks at the structure of a sentence and establishes relationships between words. This helps when trying to determine where clauses occur in the sentence. For our implementation we use Spacy's dependency parser which is an implementation of [49] to parse through each English sentence. Iterating through the tokens we check for dependency labels that signify clauses in the sentence namely -

- advcl: Adverbial clause
- ccomp: Clausal complement
- xcomp: Open clausal complement
- acl: Clausal modifier of a noun
- relcl: Relative clause

If any of these labels are present in the sentence we extract the subtree and use it to determine the clauses to apply the rules to.

Constituency parsing [49] is the second syntactic parsing method we use. Constituency parsing like dependency parsing again looks at the structure of the sentence but looks at constituents which are sub-phrases inside sentences. Using Spacy [50] and Benepar [51] libraries, we generate trees for each sentence to translate similarly to the dependency parser. Benepar uses the Berkeley Neural Parser and is commonly used in literature, so was chosen for this task.

We go through the tree and see any occurrences of the following tags in the tree:

- S: Sentence
- SBAR: Subordinating conjunction clause
- SINV: Inverted sentence
- SQ: Question sentence

We then take the leaves of each of the nodes found that are made up of these which will form our subclauses, which we can then process separately.

After using each of our techniques we have four sets of pseudo-glosses that can be used for our model but it is hard to determine which glosses are the best quality. To

determine the quality of the pseudo-glosses generated we use a qualitative and quantitative approach on a small sample of 100 lines randomly taken from the generated glosses. It is important to look at both the glosses in a quantitative and qualitative manner particular when looking at language which can be subjective with sentences being able to be translated in different ways. With multiple potential glosses being correct, having a qualitative examination helps give us confidence in the quantitative metrics. This approach was supported by a native Deaf signer who is fluent in BSL who went through each sample to manually evaluate them. The same 100 sentences were used for each sample so we can compare the performance of the glosses using only a small sample due to the time-intensive nature of the task.

Firstly, we generate our own pseudo-glosses from a native Deaf signer by hand which we can use as a ground truth. These are then able to be used to generate quantitative metrics by comparing the ground truth to the generated glosses. BLEU and ROUGE are both chosen due to their wide usage in machine translation as discussed in 3.2 which is the closest task to gloss generation. Qualitatively each sentence sampled is also judged by the native Deaf signer to be either correct, mostly correct, mostly incorrect or incorrect. We can then compare the ratio of these for each type of pseudo-gloss generated. These both give us an idea about the quality of glosses generated and enable us to compare our gloss generation methods.

We are also able to see if the glosses generated have an effect on the results of the translation model, training the model with each of the different approaches used to create the glosses. This helps to see if better pseudo-glosses help to improve the performance of the model and whether the model has any significant differences when using the different types of pseudo-glosses.

3.5 Architecture

Due to the widespread success of Transformers for translation in both traditional translation and sign language translation, this is the approach used for this paper. For the BOBSL dataset specifically we see transformers used in both [8] and [23]. With [23] getting better results we focus on how we can adapt the current architecture to utilise pseudo-glosses instead of spottings, which can be computationally expensive and rely on extra datasets.

The current architecture used in [23] as seen in Figure 2 uses 3 encoders - a context encoder, a video encoder and a spotting encoder. The spotting encoder uses spottings

obtained in the BOBSL dataset which help gain some idea of which signs are present in the data by using a dictionary dataset and looking at what signs are present. They take substantial computation to generate and are dependent on the accuracy of any spotting model and datasets available which is a limitation of the spotting encoder. Using pseudo-glosses is a more generalisable method which does not have an issue with out of vocabulary signs but does rely on the accuracy of any glosses generated.

Formally the task the transformer is aiming to complete is given an input video $V = (x_1, x_2, \dots, x_T)$ with T frames, the aim of sign language translation is to learn the conditional probability $p(S | V)$, where $S = (w_1, w_2, \dots, w_U)$ is the corresponding spoken language sentence consisting of U words [23].

In addition to the main translation task, we introduce an auxiliary objective by incorporating *pseudo-glosses*, denoted as $P_g = (g_1, g_2, \dots, g_K)$, which are annotations denoting the order of signs in the Sign Language sentence automatically generated from the target sentence S . These pseudo-glosses provide weak supervision and encourage the model to learn the intermediate structure of the sign language.

The auxiliary objective aims to model $p(P_g | V)$, learning to predict pseudo-gloss sequences directly from the video input. The encoder output associated with this auxiliary task is denoted as E_{pg} .

The final translation model conditions on both the visual input and the learned gloss representation, optimising the conditional probability:

$$p(S | V, E_{pg})$$

This multi-task setup aims to utilise the learning of the glosses to help support the I3D features and guide the learning of the translation, trying to more explicitly use the order of the signs.

The architecture proposed in this paper utilises two encoders both taking I3D features as input. One encoder being a video feature encoder and the second being a pseudo-gloss prediction encoder. This Pseudo-gloss prediction is the auxiliary task mentioned which we learn to get an understanding of the order of the signs present in the data.

Each encoder is made up of two identical layers. Our transformer encoder layer is made up of the same architecture as used in [23] using a self attention block and a position-wise feed forward network. The self-attention block first applies layer normalisation then multi-head attention then applies dropout and drop path regularisation. The feedforward network again applies normalisation and then projects through a linear layer, uses GELU activation, then dropout for regularisation and then projected through the second linear layer and again uses dropout and drop path regularisation.

The encoder output from the video embedding encoder and the pseudo-gloss encoder are both taken and fused together before being taken as input into the translation decoder. We experiment with a variety of different fusion methods in our architecture - element-wise addition, concatenation, FiLM, gated multimodal units, low-rank bilinear product and attention which we look at in detail in section 3.6 We also have a secondary decoder that is used for the auxiliary task to generate the pseudo-gloss.

Each decoder layer has the same structure each with two attention blocks, one being used for self attention and one used for cross attention with the I3D features or fused features depending on the decoder. Each block being made up of the same structure as in the encoder. We then have again have a feedforward network with the same structure as the encoder.

Each transformer is trained autoregressively producing one token at a time using the ground truth. During inference we generate tokens using greedy decoding since previous work has had similar success with greedy decoding and beam search [23]. We can see the whole architecture in FIG.

3.6 Fusion Methods

For the proposed architecture fusion of both of the encoder outputs has been done, which is then passed into the decoder for prediction of the spoken translation. To accomplish this, fusion of both of the encoded features is done. For our model we look into a variety of different fusion methods, which we can investigate via our experiments to see which fusion methods work the best for this task. For our implementation each fusion method are implemented within the model which is then trained for 20 epochs. We can then compare each of these models with the evaluation metrics ROUGE, BLEU and CHRF, which we look at in depth in section 3.2

Six different methods are investigated for this model looking at both simple and more computationally intensive methods. This can help us see how more of a factor the fusion method impacts the performance of the model and see if using a more complex method makes a significant difference.

3.6.1 Element Wise Addition

The first fusion method is element wise addition of the tensors which has been used as a baseline method other multimodal work [52]. As seen in the equation below, we simply add the elements of each tensor together. This is a basic fusion technique which we can use as a baseline to compare other fusion methods to, it has been used commonly in many multimodal architectures. It is an option which takes less computation and, so compared to other methods, could be used if the increase in accuracy is not significant. This method does not give any significant focus on either encoder and takes information from both equally.

$$\mathbf{h}_{\text{fused}} = \mathbf{h}^{(1)} + \mathbf{h}^{(2)}$$

3.6.2 Concatenation

The second method is concatenation [53], which also offers a straightforward approach to fusion. Instead of summing the tensors, we concatenate them and then pass the result through a learnable linear layer to project it back to the original d -dimensional space. While this approach introduces slightly more parameters compared to element-wise addition, it is still a relatively low-cost fusion method.

$$\mathbf{h}_{\text{fused}} = W[\mathbf{h}^{(1)} ; \mathbf{h}^{(2)}] + \mathbf{b}$$

where $[\cdot; \cdot]$ denotes concatenation.

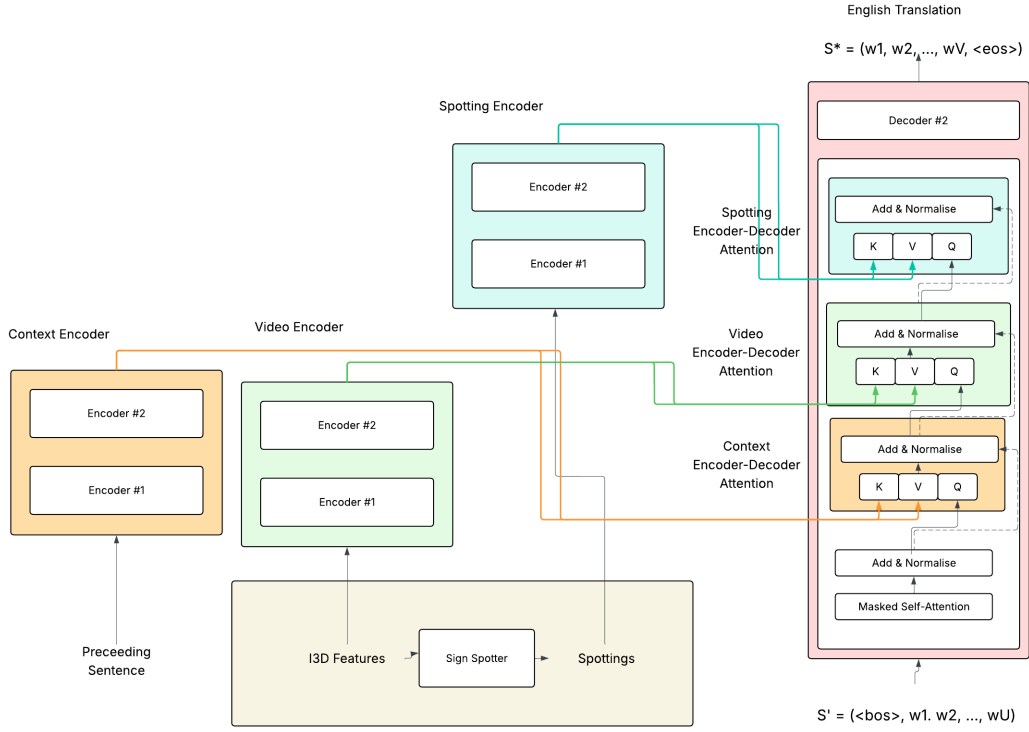


Fig. 1: Context Architecture

3.6.3 Feature-wise Linear Modulation

Using Feature-wise Linear Modulation (FiLM) [54] is a way to modulate our video encoder output using the gloss encoder output. The gloss tensor is used to generate parameters that are applied to the video encoder output. This focuses on the video encoder output which has higher significance and more reliability than the pseudo-glosses whilst using the glosses to support the tensor. The pseudo-glosses features determine how the features are scaled but are not directly fused.

$$\begin{aligned}\gamma &= W_\gamma \mathbf{h}^{(1)} + \mathbf{b}_\gamma, \\ \beta &= W_\beta \mathbf{h}^{(1)} + \mathbf{b}_\beta, \\ \mathbf{h}_{\text{fused}} &= \gamma \odot \mathbf{h}^{(2)} + \beta\end{aligned}$$

3.6.4 Gated Multimodal Units

Another approach used has been the use of gated multimodal units (GMU) [55] which is another way of being able to determine the weighting of how to use each of the encoder outputs. The GMU computes a gate which determines how much of each tensor is weighted at each step. As this is a learned feature it can dynamically prioritise the different encoders and will not weigh them equally which is potentially useful for the scenario when we have pseudo-glosses with unknown accuracy.

$$\begin{aligned}\mathbf{z} &= \sigma(W_z [\mathbf{h}^{(1)}; \mathbf{h}^{(2)}] + \mathbf{b}_z), \\ \mathbf{h}_{\text{fused}} &= \mathbf{z} \odot \mathbf{h}^{(1)} + (1 - \mathbf{z}) \odot \mathbf{h}^{(2)}\end{aligned}$$

3.6.5 Low-rank Bilinear Product

Low-rank bilinear fusion which looks at an approximation of the bilinear product [56] to look at a more expressive method of fusion compared to concatenation or additive methods. This method still does not have a significant computational cost due to using an approximation instead of the full bilinear product.

$$\begin{aligned}\mathbf{u} &= U^\top \mathbf{h}^{(1)}, \\ \mathbf{v} &= V^\top \mathbf{h}^{(2)}, \\ \mathbf{h}_{\text{fused}} &= \mathbf{u} \odot \mathbf{v}\end{aligned}$$

with $U \in R^{d_1 \times r}$, $V \in R^{d_2 \times r}$.

3.6.6 Attention for fusion

Attention [57] fusion is the final method of fusing using multi-head attention which is the most expensive method. This treats one output as the query and one as the key and value in a multihead attention mechanism. This is the most expressive method and can potentially determine complex dependencies however is the most computationally expensive method.

$$\begin{aligned}\text{Attention}(Q, K, V) &= \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V \\ Q &= H^{(1)}W^Q, \quad K = H^{(2)}W^K, \quad V = H^{(2)}W^V \\ H_{\text{fused}} &= \text{Attention}(Q, K, V)\end{aligned}$$

Using a variety of fusion methods each utilising the features in different ways, means we have a clear idea of what types of methods have an impact on translation.

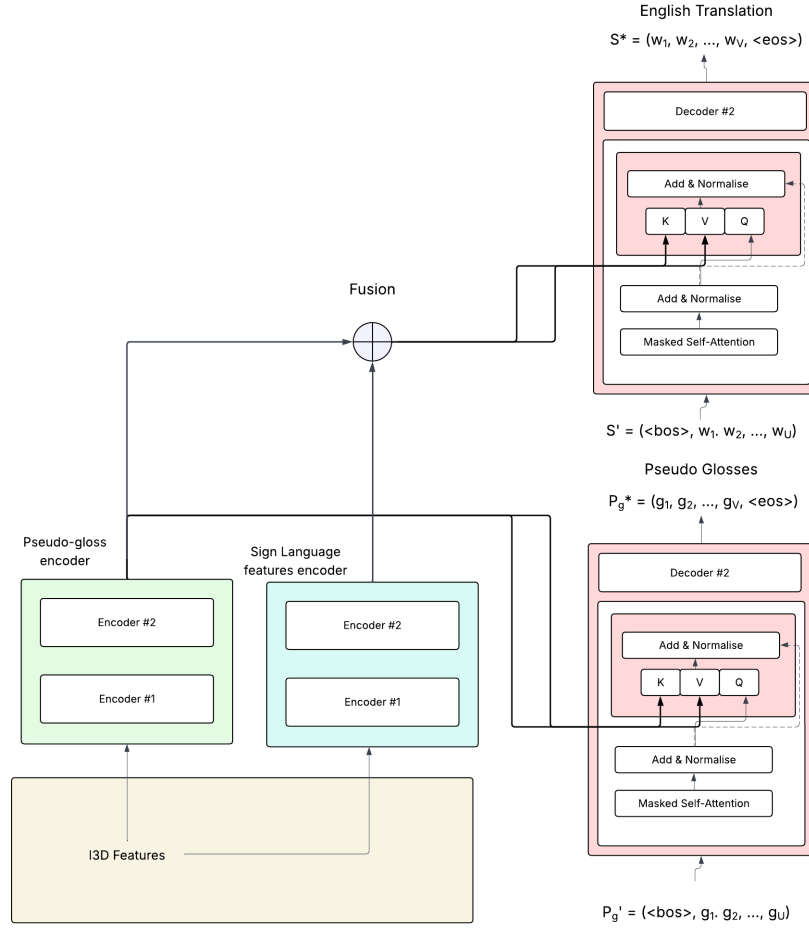


Fig. 2: Context Architecture

3.7 Implementation Considerations

When training the different models it's important to consider the amount of space and time it takes for the model to train. The models were trained on Nvidia Cuda Cluster (NCC) - a shared GPU cluster hosted in Durham, and took considerable time to train, with the original configuration scheduled to take three months. With the dataset and models taking up substantial memory using a smaller model and subset of the dataset was needed. However, using this smaller model naturally leads to worse model performance and we would not have models with similar size and dataset constraints to compare it to, so comparable models and results were needed. To achieve this, we adapted code from [23], training a scaled-down version of their model by adjusting the hyperparameters and using a reduced dataset consisting of 28 videos. The hyperparameters selected can be seen in the table 3. These results were then used as a baseline to compare our model's results to.

3.8 Hyper-Parameter Tuning

To determine the effect of hyperparameters on the performance of the model and to get the best performance out of the model we used Optuna [58] to hyperparameter tune the model.

TABLE 3: Base Context Model Hyperparameters

Hyperparameter	Value
learning rate (lr)	6e-4
weight decay (wd)	0.0
batch size (bs)	4
max epochs	30
label smoothing	0.0
lr scheduler	custom_plateau
scheduler patience	5
scheduler factor	0.7
embedding dimension (d_embedding)	256
number of encoder layers (n_layers)	2
number of attention heads (n_heads)	8
feedforward dimension (m_feedforward)	64
dropout	0.1
positional encoding	sin_cos

Optuna searches the hyperparameter space using bayesian optimisation to determine which of the hyperparameters are best. For our model we use the BLEU-4 score as the objective function since BLEU-4 is one of the hardest metric to perform well in and if the BLEU-4 score is good then you are much more likely to have a high quality translation.

We employed Optuna [?] for hyperparameter optimisation. The search space was defined as follows:

- **Embedding dimension** (`d_embedding`): chosen from {128, 192, 256}
- **Number of attention heads** (`n_heads`): selected based on `d_embedding` (e.g., ensuring divisibility)
- **Number of transformer layers** (`n_layers`): an integer in the range [2, 4]
- **Feedforward network size** (`m_feedforward`): in the range [64, 512], with a step size of 64
- **Dropout** (`dropout`): sampled uniformly from [0.0, 0.5]
- **Drop path rate** (`drop_path`): sampled uniformly from [0.0, 0.3]
- **Learning rate** (`learning_rate`): sampled log-uniformly from [1e-5, 1e-3]
- **Label smoothing** (`label_smoothing`): sampled uniformly from [0.0, 0.3]

Optuna’s Tree-structured Parzen Estimator (TPE) sampler was used for efficient exploration of the search space, and trials were evaluated using BLEU-4 score on the validation set.

These hyperparameters were chosen as they have significant impact on the performance of the model and so were prioritised for exploration.

3.9 Ethical Implications

Ethical considerations of AI models are generally important but particularly in the context of SLT models since the real-world implications are significant towards a minority community - the Deaf community whose views need to be considered. British Sign language has been historically discriminated against and is only recently been recognised as an official language of the UK, and so any approach should consider the fact its a minority language.

In terms of considerations directly linked to the language, we should consider the potential use cases of sign language models. The Deaf community are often worried about SLT models being considered as replacements for interpreters. These worries are linked with the lack of available interpreters and unqualified sign language interpreters [59] being used as cheaper replacements for qualified interpreters. Considering this the real-world applications of SLT models when performance is sufficient should be in everyday interactions and not one where you would usually have an interpreter.

Datasets also have ethical implications with their limited scope. Researchers should be aware that the datasets available tend not to have a diverse set of signers and so their models have a high chance of being affected by sampling bias. This is particularly important with BSL which has considerable linguistic variation with signs being different in different cities. Our datasets currently do not capture this linguistic variation so models created are limited in dialect and we should be clear that the models are lacking in this area.

Moreover, SLT models require full facial visibility to capture some of the non-manual features (e.g., facial expressions, mouth patterns) that are crucial to sign language. Unlike many other AI domains, data anonymisation is not feasible without significantly degrading performance. Where datasets are not publicly available, they must be

securely stored, and individuals featured in the data must be clearly informed of how their recordings may be used in both research and downstream applications. This has been done for the BOBSL dataset [7] and so is suitable for the paper.

4 RESULTS

This section outlines the different experiments and results that were conducted to investigate the optimum architecture for this translation model.

4.1 Experiments and Experimental Setting

In this study, four sets of architectural experiments were conducted: baseline model experiments, pseudo-gloss experiments, fusion experiments, and hyperparameter tuning. All experiments were performed on the NCC using either the Ampere GPU or the AA600 GPU—these were the only available GPUs with sufficient memory to accommodate both the model and the dataset. However, the limited availability of these resources constrained the amount of experimentation, as work was dependent on GPU availability.

4.2 Software Engineering Methodology

For this project, elements of an agile methodology were adopted. The work was structured around distinct objectives, with multiple iterations of the system developed, each modifying a specific component, such as a fusion method or the pseudo-gloss generation technique. This iterative approach was well-suited to the project, as it allowed for the systematic exploration of various strategies.

4.3 Pseudo-gloss Generation

As stated in Section 3.4, we carry out experiments on different methods of pseudo-gloss generation to determine which pseudo-glosses are the most accurate. These are conducted on a randomly selected subsample of 100 lines from glosses generated from 20 different files. We can see the quantitative results in Table 4 and qualitative results in Table 5.

4.3.1 Quantitative Results

After generating the pseudo-glosses using the different systems, the native Deaf signer created their own glosses using the baseline results as a guide. These were then treated as a translation, and we calculated the translation metrics for each file. As we can see in table 4 the baseline system performed significantly higher, which we believe could be due to bias created by the reference glosses being developed based on the glosses formed by the baseline system. The second-best system was using Chat GPT 4-0 mini directly, with all of the other methods having substantially lower performance.

4.3.2 Qualitative Results

Looking at the qualitative results in table 5 we can see some significant differences. The 04-mini generated pseudo-glosses have the most correct translations (30% higher than the second best method) and the lowest number of incorrect glosses. This lines up with the quantitative results and shows

TABLE 4: Evaluation Results for Pseudo-Gloss Generation

Model	ROUGE-L	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CHRF
Baseline	0.60	77.00	42.99	26.11	16.89	63.93
Dependency Parsing	0.01	0.94	0.20	0.10	0.06	9.88
Constituency Parsing	0.01	1.33	0.29	0.15	0.09	10.06
LLM Clause Splitting	0.01	6.44	0.62	0.23	0.12	10.23
LLM GPT 4o-mini Generation	0.47	28.93	12.74	4.58	1.73	40.93
LLM TinyLlama Generation	0.01	4.03	0.65	0.29	0.16	6.54

that we can be confident in the accuracy of the 04-mini pseudo-glosses. The quantitative metrics show significantly higher performance for the baseline than the qualitative metrics which we believe is due to the bias which occurred when creating them. The performance for the qualitative results should be weighed more heavily since the effect of bias is less and there is the same amount of results for each. Some of the outputs were missing, this is due to the system not coping with single word inputs and for the LLM generation, hallucination caused a significant amount of missing files when using TinyLlama.

4.4 Baseline Model

After scaling down the context-based model from [23], we were able to use these as a comparison for our own baseline model as the starting point for the work.

In table 7 we can see that we are able to get similar performance for BLEU-4, which is the main metric we want to achieve a high performance on. This shows that the model trained for a larger period of time on more data has the potential to perform well if trained to convergence.

Looking at models trained on a larger subset of data, we can see the results in table 6 outperform our baseline results by a considerable amount. This is to be expected due to the disparity in the length of training and the amount data used.

4.5 Fusion Techniques

The first set of experiments carried out was experimenting with different fusion techniques. We report both the results for the best epoch for BLEU-4 and the results at the end of 20 epochs.

Looking at the different fusion methods results in table 8 and 9, we can see that the worst performing model was the gated multimodal fusion model, having consistently low results compared to the other models. Some fusion methods performed well earlier on in training but were less consistent with attention performing well earlier on, having considerably high results for BLEU-4 and FiLM fusion also performing well but both degrading in performance. Low-rank bilinear fusion had significantly strong performance across the metrics but does degrade significantly for the BLEU-1 score in the validation subset of data. This significant drop indicates that it could be less consistent in training. Elementwise addition performed well as a baseline and performed best at its latest epoch which could lead to continued stable training.

For future work we can conclude that using either low-rank bilinear fusion or element-wise addition are good

options for fusion. Elementwise addition is potentially more stable with its best epoch being its latest epoch, whilst low-rank bilinear fusion achieves better results but is potentially less stable. Further training of both of these fusion methods is worthwhile to explore to see how they perform after training for a longer period of time on more data.

4.6 Pseudo-gloss Models

Using the baseline model which uses element-wise addition, we carried out additional experiments using the different pseudo-glosses generated. Due to time constraints only 40% of the clause splitting data was generated so we used the baseline system for any files not generated. Once again, we report both the results at the highest performing epoch and at the end of 20 epochs.

Looking at the performance for the best epoch for each model in table 10, we can see that the performance is not currently closely linked to the quality of pseudo-glosses generated based on our qualitative analysis conducted. We believe this outcome is primarily due to computational constraints, which limited both the training duration and the volume of data processed by the model. As the model was trained for only 20 epochs, it did not have enough time to fully learn the structure of the pseudo-glosses from the I3D features. Consequently, it was unable to leverage this information effectively to improve translation accuracy.

We can see that in table 11 that the directly generated pseudo-glosses are not consistently getting better and perform worse for higher epochs, showing that they could be less likely to train to convergence as quickly. The baseline and dependency pseudo-glosses perform better at higher epochs, which reflects the accuracy of the pseudo-glosses and suggests that at higher epochs, the more accurate pseudo-glosses would perform better.

4.7 Qualitative Results

We looked at the predictions for different sentences from the test set of data which you can see in tables 12,13,14 and 15 which are from the baseline models and high performing models. We can see that the models created do not have a semantic understanding of the content for most sentences predicted. Most of the sentences predicted are of a similar length, and so the model is able to learn a correlation between the signs and the length of speech.

We can see that for these samples of text the baseline model performed the best with some semantic similarity for the last prediction. The length of each sentence is also

TABLE 5: Qualitative Analysis of Pseudo-Gloss Outputs

Model	Correct	Mostly Correct	Mostly Incorrect	Incorrect	Missing
Baseline	18%	61%	15%	0%	6%
Dependency Parsing	29%	48%	20%	2%	1%
Constituency Parsing	24%	28%	40%	8%	6%
LLM Clause Splitting	35%	58%	5%	2%	0%
LLM Generation	7%	0%	0%	51%	36%
LLM Generation 04-mini	60%	38%	1%	0%	1%

TABLE 6: Impact of integrating context and spottings in video-to-text models on BOBSL.

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
<i>Video-to-Text</i>								
Albeneie et al. [3]	–	–	–	–	12.78	1.00	10.16	–
Trained with 274K	15.15	1.02	12.71	19.7	12.68	0.83	8.32	17.9
Trained with 1M	18.80	1.28	7.91	17.7	17.71	1.27	8.90	18.8
<i>Context+Video-to-Text</i>								
1 preceding sentence	20.18	1.53	9.13	18.2	19.11	1.51	9.94	19.3
2 preceding sentences	19.14	1.52	8.97	18.0	18.15	1.41	9.56	18.9
3 preceding sentences	20.05	1.56	9.08	18.1	18.82	1.48	9.64	19.1
Max 10 spottings	20.84	1.71	10.03	18.21	19.05	1.50	9.95	18.94
<i>Context+Video+Spot-to-Text</i>								
With 1 preceding sentence	25.06	2.73	11.12	22.6	24.07	2.81	12.07	23.7
With max 10 spottings	25.94	3.07	12.27	23.69	24.29	2.88	12.41	24.53

TABLE 7: Impact of integrating context and spottings in video-to-text models on BOBSL.

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
Base Context	14.08	0.21	5.50	11.64	10.32	0.11	4.31	11.67
Base Pseudo	8.89	0.23	3.56	10.06	10.77	0.13	5.17	11.30

closer, and the sentences are more readable and closer to native English. This shows promise for the baseline model to perform well but more training needs to be done to be confident in this.

Some of the predictions in table 14 are not readable and have more significant repetition between examples. This shows qualitative results between models are different, and there are clear differences in performance. This shows that the baseline model may perform better qualitatively than the low-rank bilinear fusion model despite getting worse results in most metrics.

The lack of semantic understanding and the less readable outputs are an understandable result due to the length of time training and the small subset of data the model is trained on. These are due to the computational constraints that the development environment has. With further training on a larger subset of data, the sentences could be more readable and more semantically related due to a more varied dataset and more learning.

4.8 Hyper-parameter Tuning

As discussed in Section 3.8 hyper-parameter tuning was conducted using the baseline pseudo-gloss model. This tuning uses optuna which utilises Bayesian optimisation to determine the best hyper parameters for the model. This enables higher performance by effectively searching the space for hyperparameters that enable the model to perform better. After training the model for 48 hours with each trial having a maximum of 8 epochs we were able to see how much of an impact tuning made. The chosen length of training was due to time constraints, a choice of a small maximum epoch was chosen due to this although this does limit the confidence we can have with these parameters for higher epochs. Due to time constraints we were unable to obtain the metrics for the testing subset of the data, so we have less validity in the results. This means further training needs to be done to determine the usefulness of the optimisation for our model.

We can see in table 16 the metrics achieved with the hyperparameters in table 17. Using Optuna, it is effective to find effective hyperparameters quicker than methods such

TABLE 8: Fusion methods best epoch

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
Elementwise addition	8.89	0.23	3.56	10.06	10.77	0.13	5.17	11.30
Concatenation	10.90	0.32	5.09	10.70	6.77	0.17	4.68	10.58
FiLM	10.86	0.34	4.20	12.23	12.37	0.18	5.05	14.19
Gated Multimodal Units	6.99	0.24	3.41	8.78	7.33	0.08	4.39	8.92
Low-rank Bilinear	11.53	0.33	4.80	11.30	11.31	0.54	4.97	11.86
Attention	12.70	0.35	5.44	10.28	12.20	0.29	5.69	9.83

TABLE 9: Fusion methods 20 epochs

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
Elementwise addition	8.89	0.23	3.56	10.06	10.77	0.13	5.17	11.30
Concatenation	10.09	0.11	4.18	10.84	11.27	0.22	4.69	11.49
FiLM	7.75	0.10	3.49	10.23	10.03	0.16	5.34	11.96
Gated Multimodal Units	6.99	0.24	3.41	8.78	7.33	0.08	4.39	8.92
Low-rank Bilinear	9.58	0.17	4.38	10.16	10.72	0.26	5.84	11.00
Attention	7.84	0.14	3.66	6.86	7.34	0.14	4.50	6.52

TABLE 10: Pseudo-gloss methods best epoch

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
Baseline Pseudo-gloss	8.89	0.23	3.56	10.06	10.77	0.13	5.17	11.30
Dependency Pseudo-gloss	7.80	0.21	3.05	9.06	10.02	0.21	3.88	10.78
Constituency Pseudo-gloss	5.94	0.14	3.79	9.78	7.75	0.15	4.90	10.81
LLM clause segmentation Pseudo-gloss	10.25	0.21	4.21	9.79	10.51	0.24	4.71	10.10
LLM generation Pseudo-gloss	9.48	0.26	4.25	10.14	10.98	0.17	5.37	11.44

TABLE 11: Pseudo-gloss methods 20 epochs

Model	Val				Test			
	BLEU-1	BLEU-4	ROUGE	CHRF	BLEU-1	BLEU-4	ROUGE	CHRF
Baseline Pseudo-gloss	8.89	0.23	3.56	10.06	10.77	0.13	5.17	11.30
Dependency Pseudo-gloss	7.80	0.21	3.05	9.06	10.02	0.21	3.88	10.78
Constituency Pseudo-gloss	6.90	0.06	3.08	9.02	8.78	0.17	4.62	10.24
LLM clause segmentation Pseudo-gloss	6.30	0.05	2.28	8.98	9.95	0.16	4.51	11.33
LLM generation Pseudo-gloss	7.20	0.10	3.20	9.14	9.98	0.13	4.73	11.45

as grid search. With the hyperparameter search we achieved better results with BLEU-1 than the baseline, but got lower BLEU-4 values. This shows that the hyperparameter tuning may not have a significant effect. Further tuning with a longer epoch time could make a difference since each trial was only run for 8 epochs, however, this was not possible with the time constraints.

5 EVALUATION

This section looks at how the different components of the system perform and how they could be improved and explored further.

5.1 Pseudo-Glosses

Comparing the base pseudo-gloss generation system to the four different improved methods, half of the methods implemented were an improvement to the base model, with both dependency parsing and LLM clause splitting improving the accuracy of the samples generated.

Constituency parsing was not an improvement from the sample generated and in fact had worse accuracy than the base rule based model. This could be due to the sample chosen, since only 100 sentences were used for each of the methods. The parsing method may be splitting too often, which could be the cause of the decrease in accuracy. How the sentence is split affects how the rule-based system is applied, and with further adjustments to the amount of

TABLE 12: Example outputs from the context model on different inputs

Generated Caption (Baseline)	Video Description (Reference)
Yeah. A peninsula bursting with wildlife and breathtaking views, on a coastline stuffed with stories.	OK. It's a very interesting thing, because the most of the social occasion.
A treat for the eyes and the taste buds. This is enormous.	It's a bit of fun. It's a wonderful lovely night,

TABLE 13: Example outputs from the baseline pseudo-gloss model on different inputs

Generated Caption (Baseline)	Video Description (Reference)
Yeah? A peninsula bursting with wildlife and breathtaking views, on a coastline stuffed with stories.	OK. It's a very big job to see the whole thing, and it's not just a very nice.
A treat for the eyes and the taste buds. This is enormous.	I'll have a look at that. It's a big

TABLE 14: Example outputs from the Low-rank Bilinear Pseudo-gloss model on different inputs

Generated Caption (Baseline)	Video Description (Reference)
Yeah. A peninsula bursting with wildlife and breathtaking views, on a coastline stuffed with stories.	OK. And I'm not sure that I'm not taking my way to the moment, but I'm not going to be able to be able to be able to
A treat for the eyes and the taste buds.	I'm not going to be at the moment.
This is enormous.	I'm not going to be at the moment

TABLE 15: Example outputs from the Dependency Pseudo-gloss model on different inputs

Generated Caption (Baseline)	Video Description (Reference)
Yeah. A peninsula bursting with wildlife and breathtaking views, on a coastline stuffed with stories.	OK. And it was a very difficult to be in the same way of the unit.
A treat for the eyes and the taste buds.	There we are.
This is enormous.	And it's

TABLE 16: Optuna Results

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE
Optuna	14.17	3.06	0.69	0.199	6.78

TABLE 17: Optuna Hyperparameters

Parameter	Value
d_embedding	190
n_heads	6
n_layers	3
m_feedforward	380
dropout	0.25
drop_path	0.12
learning_rate	5.8e-4
label_smoothing	0.045

splitting could lead to increased performance.

The second method, which had substantially decreased accuracy, was using the TinyLlama LLM to generate the pseudo-glosses directly. The selection of LLM was due to time constraints and computing resources. Using a smaller

LLM was necessary to generate the glosses in an acceptable time. This naturally lead to decreased accuracy with issues with hallucination, causing some outputs not to have any of the original sentence or some sentences being completely unchanged. We know that the failures of this method were due to the LLM chosen since another experiment was done using a larger LLM which had the highest qualitative results out of the 6 different samples. This shows that using LLMs to generate pseudo-glosses is the most successful method explored and is recommended if they have access to a more complex model. Otherwise, using dependency parsing in conjunction with the rule-based system is best.

Utilising rule-based system also has significantly less computation time and resources which is a big advantage when one of the primary motivations for the use of these pseudo-glosses is to decrease the need for computation like sign spotting or mouth spottings. Whilst using larger LLMs does increase accuracy, this also increase the amount of resources needed. Using sign spottings and mouth spottings could be a better technique to use in that scenario if it is less computationally intensive than the LLM. One scenario where LLMs could be particularly valuable is in translating

sign languages for which accurate dictionary datasets or mouth-spotting patterns are unavailable. In such cases, an LLM only requires knowledge of the approximate word order of the sign language, allowing it to generate glosses with high accuracy.

One significant way to improve this work would be to collaborate with BSL linguists and improve current rule-based systems. The current system this paper builds on was not created by linguists and by a student with some advice from a Deaf signer. Having expertise with linguistics could substantially increase the robustness of the rules created which would have a significant impact on the accuracy of pseudo-glosses generated. A major drawback of [23] is the lack of significant examination by Deaf researchers, especially since the study is situated within the domain of computer science rather than Deaf studies. As a result, it has not undergone the necessary scrutiny from those familiar with the linguistics of BSL. [23] is currently the only recent paper looking at rule-based systems for BSL, and so was selected, but if other research is conducted looking at further papers would help improve the validity of the pseudo-gloss generation.

There are limitations to the validity of the pseudo-glosses generated as we do not have the resources to hire annotators to annotate the original data to compare our pseudo-glosses to accurate glosses. Still, the validity of the actual pseudo-glosses is not as important if the overall performance of the model improves.

5.2 Embeddings

The two different ways embeddings are used for our work is for textual embeddings and sign language feature embeddings. The textual embeddings are using BERT-base case and this method is commonly used for language models. This embedding is suitable for the context and experimenting further with textual embeddings would likely not lead to any significant increase in performance. Where there is a difference in embeddings compared to some SLT work is in the sign language feature embeddings.

The current use of embeddings is similar to previous work [7], [23] and therefore has validity in its effectiveness, but further exploration of embeddings could have a significant effect on the translation accuracy and could be worth exploring in future work to see the effect of this on the accuracy.

One of the best performing models, TwoStream-SLT [10] for the Phoenix14T dataset uses frames as the key information along with keypoint estimation, using HRNet, to supplement information to the frames. This keypoint estimation focuses on the body, hands and face. This approach takes into account the blur in the frames, which can impact the performance of keypoint estimation. This approach uses it as supplemental information instead of the primary encoding. Building upon the TwoStream architecture, we could use optical flow through an additional encoder and explore the effects of this on performance.

Overall, the use of embeddings is appropriate for the task and is useful in directly determining the effect of pseudo-glosses since the use of embeddings is similar to previous work. Utilising optical flow or using other architectures to generate embeddings from the sign language

videos is where this could be further explored and increase the performance of the model.

5.3 Model Architecture

Looking at our model architecture, it manages to get comparable results for a subset of the data to the [23] model when the model is scaled down to the same size. This shows that given more resources, there is potential that the model could train to convergence and produce comparable translations. Currently it is hard to determine whether the model would have comparable results for a larger subset of data for certain but the work done so far is a good indication that the model architecture would perform satisfactory.

With more computational resources, more experimentation could be done for longer epochs with a larger model size which would enable us to determine whether our assumptions are correct and be able to compare our results with their trained model.

Our architecture is building upon the traditional encoder-decoder architecture and so has a strong foundation, which means we can be confident that the architecture design would work well for translation. The exact usage of two encoder-decoder works for the use of an auxiliary task which is the main focus of the design of the architecture, and so the choice of architecture was effective.

To build upon this architecture similarly to how [23] uses multiple encoders we could extend our architecture. Context is used effectively in their model and substantially increased performance, and so could help performance and get even more competitive results than the current model. We could also incorporate other encoders into our architecture using optical flow which lets the model learn from optical flow explicitly rather than solely using the I3D model which uses the optical flow of the video.

5.4 Dataset Limitations

Most of the limitations to our model stems from the source of our data. Using the BOBSL dataset whilst still the best option available leads to many different limitations. The dataset will contain less regional variation than is present in the real-world with interpreters tending to use signs that are more well known. It contains limited variation in the demographic of the signers due to all of the signers being interpreters, limited numbers of signers and it lacks organic sign language. These limitations mean that the model trained on this data is likely to perform poorly when faced with variations in signing styles not represented in the dataset, reducing its applicability in real-world scenarios.

The methodology of collection of the dataset means that all of the data is filmed in similar conditions which means that the model will suffer in conditions that are dissimilar to the studio conditions. Different lighting could significantly impact the performance of the model so again would lead to the model being unable to perform in real-world conditions.

These limitations are not large considerations as the methods for pseudo-gloss generation if successful can be generalised to future more representative datasets as long as there is a spoken language translation for the data which can improve the performance of future models.

5.5 Deaf Involvement

One aspect this paper lacks is Deaf involvement, which is emphasised in interdisciplinary papers regarding Sign Language Technology [6]. Whilst the paper has an informed perspective as the author has grown up in the Deaf Community and the ideas proposed in the paper have been discussed with native Deaf Signer, this has been limited. This is a natural byproduct of this report being done by one hearing person but still needs to be acknowledged. If Deaf academics are collobaorted with for future work this would ensure that the perspective of the Deaf community is included and could help bring to light other issues that the author may not be aware of. This could particularly, be helpful for the pseudo-gloss generation as previously stated as the use of BSL linguistics could be a significant help in improving these systems.

This paper manages to consider the Deaf perspective looking at works done by Deaf people that are linked with the paper which is a significant difference to a lot of work which often are uninformed or make incorrect assumptions of the language. For the pseudo-gloss generation whilst there was no Deaf people involved in the creation of the systems a native Deaf signer was used to evaluate the system which is important to help ensure the validity of the pseudo-gloss generation.

6 CONCLUSION

In this section we look at how we meet the objectives of the work, how this work could be applied in the literature and where this work could be extended.

6.1 Objectives

The work done manages to meet our aim of improving pseudo-gloss generation with reasonable confidence for BSL. We manage to propose four different methods that extend the current rule-based system proposed in [37] with three of these methods showing promise for future use. The use of LLM for this task is a significant contribution and one which greatly improves on rule-based systems or other methods where we do not have the datasets to learn the association between annotated glosses and their spoken language translation.

Another focus is an exploration of the architecture and the usage of pseudo-glosses within this architecture. Within this work, we manage to propose our own design for the architecture building from the traditional transformer-decoder. The use of pseudo-gloss generation as an auxiliary task to improve translation is a significant contribution, which we was another aim.

Both of these aspects address our intermediate and basic deliverables. However, the advanced deliverables were not fully met due to the time constraints imposed by the computational limitations of the model.

6.2 Applications

Each major contribution can be applied to help improve SLT models.

The use of pseudo-gloss generation can be used for application for any new datasets that are created particularly

those which cover the issues that current datasets have. The various methods can be used in different scenarios. Using the rule-based system extended with dependency parsing can improve the results for a scenario where they are needed to be generated where we have a lack of computational resources with the drawback that a rule-based system needs to be created for the target sign language. In the scenario where we have a target language and we do not have the resources or knowledge to create a rule-based system for the language we can use the LLM approach to directly generate the pseudo-glosses which is a distinct advantage for low resource languages. This is particularly useful if we have datasets that do not have spotting datasets or mouth spotting models.

The use of pseudo-gloss generation as an auxiliary task could be used as an extension for many existing SLT transformer models as long as the architecture is reworked to include the encoder output in generation. Whilst it is not certain how much of an impact this could have it is a worthwhile exploration to increase accuracy. When the architecture is trained this could also be used to help develop gloss annotations across datasets in the case that the accuracy is satisfactory on a diverse set of data. This is useful if we have sign language data that do not contain the spoken language translations since all of the current methods of gloss generation require this.

6.3 Future Work

Building from the work done in this paper there are different avenues for future exploration.

With more time or computational resources, further training of the models for a longer period could help produce results with higher validity where we could be more certain of the implications of the use of pseudo-glosses. Likewise, using a larger subsection of the data for training and testing would also allow us to compare the results to the models in the literature on the same test set.

Outside of future training there are other avenues for exploration to increase performance. As stated in the evaluation we can look at the embeddings and experiment with utilising different ones than are typically used in BOBSL models. Collaboration with sign language linguists could potentially be the most impactful in the development of rule-based pseduo-gloss generation since the robustness of the rules used has a significant impact for the accuracy of the rule-based systems.

While this work improves the accuracy of pseudo-glosses by using clause segmentation, this can only go so far as long as the rule-based system remains imperfect. The fact that the current system has not been reviewed by linguists means that we cannot be certain of its accuracy. This represents a significant area for future work, where further refinement and validation could greatly enhance the system's accuracy and validity.

REFERENCES

- [1] Kayo Yin, Amit Moryossef, Julie Hochgesang, Yoav Goldberg, and Malihe Alikhani. Including signed languages in natural language processing. *arXiv preprint arXiv:2105.05222*, 2021.
- [2] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Neural sign language translation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7784–7793, 2018.
- [3] Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10023–10033, 2020.
- [4] Alan Davies and Catherine Elder. *The handbook of applied linguistics*. Wiley Online Library, 2004.
- [5] Rachel Sutton-Spence and Bencie Woll. *The linguistics of British Sign Language: an introduction*. Cambridge University Press, 1999.
- [6] Neil Fox, Bencie Woll, and Kearsy Cormier. Best practices for sign language technology research. *Universal Access in the Information Society*, pages 1–9, 2023.
- [7] Samuel Albanie, Gül Varol, Liliane Momeni, Hannah Bull, Triantafyllos Afouras, Himel Chowdhury, Neil Fox, Bencie Woll, Rob Cooper, Andrew McParland, et al. Bbc-oxford british sign language dataset. *arXiv preprint arXiv:2111.03635*, 2021.
- [8] Samuel Albanie, Gül Varol, Liliane Momeni, Triantafyllos Afouras, Joon Son Chung, Neil Fox, and Andrew Zisserman. Bsl-1k: Scaling up co-articulated sign language recognition using mouthing cues. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XI 16*, pages 35–53. Springer, 2020.
- [9] Amit Moryossef, Kayo Yin, Graham Neubig, and Yoav Goldberg. Data augmentation for sign language gloss translation. *arXiv preprint arXiv:2105.07476*, 2021.
- [10] Yutong Chen, Ronglai Zuo, Fangyun Wei, Yu Wu, Shujie Liu, and Brian Mak. Two-stream network for sign language recognition and translation. *Advances in Neural Information Processing Systems*, 35:17043–17056, 2022.
- [11] Jiangbin Zheng, Yidong Chen, Chong Wu, Xiaodong Shi, and Suhail Muhammad Kamal. Enhancing neural sign language translation by highlighting the facial expression information. *Neurocomputing*, 464:462–472, 2021.
- [12] Shiwei Gan, Yafeng Yin, Zhiwei Jiang, Lei Xie, and Sanglu Lu. Skeleton-aware neural sign language translation. In *Proceedings of the 29th ACM International Conference on Multimedia*, pages 4353–4361, 2021.
- [13] Kayo Yin and Jesse Read. Better sign language translation with stmc-transformer. *arXiv preprint arXiv:2004.00588*, 2020.
- [14] Kearsy Cormier, Adam Schembri, and Bencie Woll. Pronouns and pointing in sign languages. *Lingua*, 137:230–247, 2013.
- [15] Adam Schembri, Rose Stamp, Jordan Fenlon, and Kearsy Cormier. Variation and change in varieties of british sign language in england. *Sociolinguistics in England*, pages 165–188, 2018.
- [16] Katherine Rowley and Kearsy Cormier. Attitudes towards age variation and language change in the british deaf community. *Language & Communication*, 92:15–32, 2023.
- [17] Katherine Rowley and Kearsy Cormier. Accent or not? language attitudes towards regional variation in british sign language. *Applied Linguistics Review*, 14(4):919–943, 2023.
- [18] Patricia Cabot Álvarez. Sign language translation with pseudo-glosses. B.S. thesis, Universitat Politècnica de Catalunya, 2022.
- [19] Ru Peng, Yawen Zeng, and Junbo Zhao. Better sign language translation with monolingual data. *arXiv preprint arXiv:2304.10844*, 2023.
- [20] Danilo Avola, Marco Bernardi, Luigi Cinque, Gian Luca Foresti, and Cristiano Massaroni. Exploiting recurrent neural networks and leap motion controller for the recognition of sign language and semaphoric hand gestures. *IEEE Transactions on Multimedia*, 21(1):234–245, 2018.
- [21] Anshul Mittal, Pradeep Kumar, Partha Pratim Roy, Raman Balasubramanian, and Bidyut B Chaudhuri. A modified lstm model for continuous sign language recognition using leap motion. *IEEE Sensors Journal*, 19(16):7056–7063, 2019.
- [22] Paul D Rosero-Montalvo, Pamela Godoy-Trujillo, Edison Flores-Bosmediano, Jorge Carrascal-García, Santiago Otero-Potosi, Henry Benitez-Pereira, and Diego H Peluffo-Ordóñez. Sign language recognition based on intelligent glove using machine learning techniques. In *2018 IEEE Third Ecuador Technical Chapters Meeting (ETCM)*, pages 1–5. IEEE, 2018.
- [23] Ozge Mercanoglu Sincan, Necati Cihan Camgoz, and Richard Bowden. Is context all you need? scaling neural sign language translation to large domains of discourse. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1955–1965, 2023.
- [24] Angus Grieve-Smith. Angus grieve-smith: 10 reasons why sign-to-speech technology won’t be practical anytime soon, May 2016.
- [25] Danielle Bragg, Oscar Koller, Mary Bellard, Larwan Berke, Patrick Boudreault, Annelies Braffort, Naomi Caselli, Matt Huenerfauth, Hernisa Kacorri, Tessa Verhoef, et al. Sign language recognition, generation, and translation: An interdisciplinary perspective. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*, pages 16–31, 2019.
- [26] Bowen Shi, Diane Brentari, Greg Shakhnarovich, and Karen Livescu. Open-domain sign language translation learned from online video. *arXiv preprint arXiv:2205.12870*, 2022.
- [27] Yutong Chen, Fangyun Wei, Xiao Sun, Zhirong Wu, and Stephen Lin. A simple multi-modality transfer learning baseline for sign language translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5120–5130, 2022.
- [28] Junfu Pu, Wengang Zhou, and Houqiang Li. Iterative alignment network for continuous sign language recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4165–4174, 2019.
- [29] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, and Richard Bowden. Subunets: End-to-end hand shape and continuous sign language recognition. In *Proceedings of the IEEE international conference on computer vision*, pages 3056–3065, 2017.
- [30] Zhe Niu and Brian Mak. Stochastic fine-grained labeling of multi-state sign glosses for continuous sign language recognition. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI 16*, pages 172–186. Springer, 2020.
- [31] Runkeng Cui, Hu Liu, and Changshui Zhang. A deep neural framework for continuous sign language recognition by iterative training. *IEEE Transactions on Multimedia*, 21(7):1880–1891, 2019.
- [32] Hao Zhou, Wengang Zhou, Yun Zhou, and Houqiang Li. Spatial-temporal multi-cue network for sign language recognition and translation. *IEEE Transactions on Multimedia*, 24:768–779, 2021.
- [33] Yong Cao, Wei Li, Xianzhi Li, Min Chen, Guangyong Chen, Long Hu, Zhengdao Li, and Hwang Kai. Explore more guidance: A task-aware instruction network for sign language translation enhanced with data augmentation. *arXiv preprint arXiv:2204.05953*, 2022.
- [34] Sang-Ki Ko, Chang Jo Kim, Hyedong Jung, and Choongsang Cho. Neural sign language translation based on human keypoint estimation. *Applied sciences*, 9(13):2683, 2019.
- [35] Aoxiong Yin, Tianyun Zhong, Li Tang, Weike Jin, Tao Jin, and Zhou Zhao. Gloss attention for gloss-free sign language translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2551–2562, 2023.
- [36] Zhengsheng Guo, Zhiwei He, Wenxiang Jiao, Xing Wang, Rui Wang, Kehai Chen, Zhaopeng Tu, Yong Xu, and Min Zhang. Unsupervised sign language translation and generation. *arXiv preprint arXiv:2402.07726*, 2024.
- [37] Phoebe Alexandra Pinney and Riza Theresa Batista-Navarro. English2bsl: A rule-based system for translating english into british sign language. In *Proceedings of the Third Workshop on Language Technology for Equality, Diversity and Inclusion*, pages 10–16, 2023.
- [38] Adam Schembri, Jordan Fenlon, Ramas Rentelis, Sally Reynolds, and Kearsy Cormier. Building the british sign language corpus. 2013.
- [39] Samuel Albanie, Gül Varol, Liliane Momeni, Triantafyllos Afouras, Andrew Brown, Chuhan Zhang, Ernesto Coto, Necati Cihan Camgoz, Ben Saunders, Abhishek Dutta, et al. Seehear: Signer diarisation and a new dataset. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 2280–2284. IEEE, 2021.
- [40] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.
- [41] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.
- [42] Maja Popović. chrF: character n-gram f-score for automatic mt

- evaluation. In *Proceedings of the tenth workshop on statistical machine translation*, pages 392–395, 2015.
- [43] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset, 2018.
 - [44] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
 - [45] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush. Transformers: State-of-the-art natural language processing. In Qun Liu and David Schlangen, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online, October 2020. Association for Computational Linguistics.
 - [46] Markus Frohmann, Igor Sterner, Ivan Vulić, Benjamin Minixhofer, and Markus Schedl. Segment any text: A universal approach for robust, efficient and adaptable sentence segmentation, 2024.
 - [47] Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. Tinyllama: An open-source small language model, 2024.
 - [48] OpenAI. Chatgpt (gpt-4o mini version), 2025. Large language model.
 - [49] Matthew Honnibal and Mark Johnson. An improved non-monotonic transition system for dependency parsing. In Lluís Màrquez, Chris Callison-Burch, and Jian Su, editors, *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 1373–1378, Lisbon, Portugal, September 2015. Association for Computational Linguistics.
 - [50] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. spacy: Industrial-strength natural language processing in python. 2020.
 - [51] Nikita Kitaev and Dan Klein. Constituency parsing with a self-attentive encoder. In Iryna Gurevych and Yusuke Miyao, editors, *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2676–2686, Melbourne, Australia, July 2018. Association for Computational Linguistics.
 - [52] Yahya Massoud and Robert Laganieri. Learnable fusion mechanisms for multimodal object detection in autonomous vehicles. *IET Computer Vision*, 18(4):499–511, 2024.
 - [53] Guy D. Rosin, Eytan Adar, and Kira Radinsky. Learning word relatedness over time. In Martha Palmer, Rebecca Hwa, and Sebastian Riedel, editors, *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 1168–1178, Copenhagen, Denmark, September 2017. Association for Computational Linguistics.
 - [54] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer, 2017.
 - [55] John Arevalo, Tamar Solorio, Manuel Montes y Gómez, and Fabio A. González. Gated multimodal units for information fusion, 2017.
 - [56] Jin-Hwa Kim, Kyoung-Woon On, Woosang Lim, Jeonghee Kim, Jung-Woo Ha, and Byoung-Tak Zhang. Hadamard product for low-rank bilinear pooling, 2017.
 - [57] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023.
 - [58] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework, 2019.
 - [59] Elizabeth Schniedewind, Ryan Lindsay, and Steven Snow. Ask and ye shall not receive: Interpreter-related access barriers reported by deaf users of american sign language. *Disability and Health Journal*, 13(4):100932, 2020.